

MAT 330

Chapter 2

2. Function composition of polynomials with real coefficients;
Multiplication of 2×2 matrices with integer entries

3. None

4. $\{0, 4, 8, 12\}$ under $\oplus \text{ mod } 16$

For every a, b in set S , $a \cdot b \in S$

$\{1, 4, 9, 13\}$ under $\otimes \text{ mod } 15$

For every a, b in set S , $a \cdot b \in S$

5. $T_n \mathbb{Z}_{20}, 13^{-1} = n - 13$
 $= 7$

b. By trial and error

$$13^2 \text{ mod } 14 \equiv 13 \cdot 13 \text{ mod } 14$$

$$= 169 \text{ mod } 14$$

$$= 1$$

$\therefore 13$ is its own inverse in $U(14)$

c. Find b in $U(n)$ such that $(n-1) \cdot b \equiv 1 \text{ mod } n$ — (1)

Notice $n-1 \equiv -1 \text{ mod } n$

From (1) $-1 \cdot b \equiv 1 \text{ mod } n$

$$-b \equiv 1 \text{ mod } n$$

$$b \equiv -1 \equiv n-1 \text{ (mod } n)$$

$\therefore$ Inverse of $(n-1)$ in $U(n)$ is itself since

$$(n-1) \cdot (n-1) = n^2 - 1$$

$$(n^2 - 1) \text{ mod } n \equiv n^2 \text{ mod } n + 1 \text{ mod } n \equiv 1 \text{ (identity element)}$$

d. Let z be a general complex number in the complex numbers set with form $z = a + bi$

$$\text{From surds } z^{-1} = \frac{1}{z} = \frac{1}{a+bi} \cdot \frac{a-bi}{a-bi} = \frac{a-bi}{a^2+b^2} \quad (i^2 = -1)$$

$$3-2i$$

$$(3-2i)^{-1} = \frac{3+2i}{(3-2i)(3+2i)} = \frac{3+2i}{13}$$

$$\begin{aligned} \text{6a. } (7+5i)(-3+2i) &= -21 + 14i - 15i + 10i^2 \\ &= -21 - 10 - i \\ &= -31 - i \end{aligned}$$

$$\text{b. } \begin{vmatrix} 7 & 4 \\ 1 & 5 \end{vmatrix} = (7 \cdot 5 - 4 \cdot 1) \bmod 13 = 31 \bmod 13 = 5$$

$$\text{c. } \begin{bmatrix} 6 & 3 \\ 8 & 2 \end{bmatrix}^{-1} = \frac{1}{6 \cdot 2 - 8 \cdot 3} \begin{bmatrix} 2 & -3 \\ -8 & 6 \end{bmatrix} = \frac{1}{-12} \begin{bmatrix} -2 & 3 \\ 8 & -6 \end{bmatrix}$$

$$\text{d. } \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}^{-1} = \frac{1}{(2 \cdot 3 - 1 \cdot 1) \bmod 7} \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}$$

$$\text{8. } U(20) = \{1, 3, 7, 9, 11, 13, 17, 19\}$$

11. Inverse of a such that $a \cdot a^{-1} = e$

$$\text{We know } a^{12} = e$$

$$a^{-1} = a^{12-1} = a^{11}$$

ii Find $(a^4)^{-1}$ such that $a^4 \cdot (a^4)^{-1} = e$

$$(a^4)^{-1} = a^{12-4} = a^8$$

$\therefore a^4$ is its inverse

iii Find $(a^2)^{-1}$ such that $a^2 \cdot (a^2)^{-1} = e$

$$(a^2)^{-1} = a^{12-2} = a^{10}$$

iv Find $(a^{11})^{-1}$ such that $a^{11} \cdot (a^{11})^{-1} = e$

$$(a^{11})^{-1} = a^{12-11} = a$$

$$\text{12. } U(9) = \{1, 2, 4, 5, 7, 8\}$$

For $2, 7, 8$ in $U(9)$, find x such that $a \cdot x \bmod 9 = 1$

$$2 \cdot 5 \bmod 9 \equiv 10 \bmod 9 = 1$$

$$\therefore 2^{-1} = 5$$

$$\text{v. } 7 \cdot 4 \bmod 9 \equiv 28 \bmod 9 = 1$$

$$\therefore -1^{-1} = -1$$

$$\text{iii } b \equiv n-1$$

In any $U(n)$ group, $n-1$ inverse element is itself

$$\therefore b^{-1} = b$$

16.	⊗ ₁₀	5	15	25	35	Using a simple Cayley table, we've shown the binary operation on this subset of $\mathbb{Z}_{40}$ satisfies the condition for a group
	5	25	35	5	15	
	15	35	25	15	5	
	25	5	15	25	35	
	35	15	5	35	25	

Identity element is 25

They both exhibit order of 4

Both are under similar operations of modular multiplication

17 To show $G = H$, prove $G \subseteq H$ and $H \subseteq G$

i Let $x \in G$. Since G is a group we know x^{-1} is in G .
Now x^{-1} is in H by its definition.

$\therefore \forall x \in G$, it follows that $\exists x^{-1} \in H \quad G \subseteq H$

Let $y \in H$. By definition, $\exists x \in G$ such that $y = x^{-1}$.

Since $x \in G$ and G is a group it follows that $x^{-1} \in G$.

However, $y = x^{-1}$, so $y \in G$.

$\therefore H \subseteq G$

Since $G \subseteq H$ and $H \subseteq G$, G must be equal to H .

20 Any element $x \in U(n)$ satisfies $x^2 \equiv 1 \pmod{n}$ iff
 $x^2 - 1 \equiv 0 \pmod{n}$

$$\Rightarrow (x-1)(x+1) \equiv 0 \pmod{n}$$

Check for 2-case solutions $x=1$; $x=-1$

$$x=1$$

$$1^2 \equiv 1 \pmod{n}$$

Hence, $x=1$ is a solution and $1 \in U(n)$ since 1 is always coprime

ii. $x = -1$

$$(-1)^n \equiv 1 \pmod{n}$$

Hence, $x = -1$ is also a solution but not in $U(n)$.

However, we can rewrite $-1 \equiv (n-1) \pmod{n}$, we represent the solution with element $n-1$.

Since $1 \neq n-1$ for $n > 2$, we've proved there will at least be these 2 elements in $U(n)$ that will satisfy this condition.

25. To prove the overall statement, we need to prove both implications. Statement here:

- i. If G is abelian then $(ab)^{-1} = a^{-1}b^{-1} \forall a, b \in G$
- ii. If $(ab)^{-1} = a^{-1}b^{-1} \forall a, b \in G$, then G is abelian.

i. Let G be an abelian group

By property of inverses, we know $(ab)(ab)^{-1} = e$

$$a^{-1} \cdot (ab)(ab)^{-1} = a^{-1} \cdot e$$

$$(a^{-1} \cdot a) b (ab)^{-1} = a^{-1}$$

$$e b (ab)^{-1} = a^{-1}$$

$$b \cdot b (ab)^{-1} = b^{-1} \cdot a^{-1}$$

$$(b^{-1} \cdot b) (ab)^{-1} = b^{-1} \cdot a^{-1}$$

$$(ab)^{-1} = b^{-1} \cdot a^{-1}$$

$\therefore$ If G is abelian $(ab)^{-1} = a^{-1} \cdot b^{-1}$

ii. Let $(ab)^{-1} = a^{-1} \cdot b^{-1} \forall a, b \in G$ [show G is abelian]

$$(ab)^{-1} = a^{-1} \cdot b^{-1}$$

$$((ab)^{-1})^{-1} = (a^{-1} \cdot b^{-1})^{-1}$$

$$ab = (a^{-1} \cdot b^{-1})^{-1}$$

Using property of inverses

$$(a^{-1} \cdot b^{-1})^{-1} = (b^{-1})^{-1} \cdot (a^{-1})^{-1} = ba$$

$$\therefore ab = ba \quad \forall a, b \in G$$

G must be abelian

Thus G is abelian iff $(ab)^{-1} = a^{-1} \cdot b^{-1} \forall a, b \in G$

$$30 \quad |Z_{105}| = 105$$

$$|Z_{44}| = 44$$

$$|D_{22}| = 44$$

$$33 \quad \begin{array}{c|cccc} & e & a & b & c & d \\ \hline e & e & a & b & c & d \\ a & a & b & c & d & e \\ b & b & c & d & e & a \\ c & c & d & e & a & b \\ d & d & e & a & b & c \end{array}$$

34 Like in #25, we need to prove it both ways

$$\therefore \text{If } ab=ba \text{ then } (ab)^2 = a^2b^2$$

$$\therefore \text{If } (ab)^2 = a^2b^2 \text{ then } ab=ba$$

$$i. \text{ Let } ab=ba$$

$$(ab)^2 = (ab)(ab)$$

$$(ab)(ab) = a(ba)b$$

$$a(ba)b = a(ab)b = aabb = a^2b^2$$

$$\therefore \text{ Let } (ab)^2 = a^2b^2$$

$$(ab)^2 = (ab) \cdot (ab) = ab \cdot ab$$

$$a^2b^2 = (aa)(bb) = aabb$$

$$ab \cdot ab = aabb$$

$$ab \cdot ab - aabb = 0$$

$$ab \cdot ab - aabb = a(bab - abb) = 0$$

$$= a(ba - ab)b = 0$$

Since $a, b \in G$, $ba - ab$ must equal 0

$$ba - ab = 0$$

$$ab = ba$$

If $ab = ba$ then $(ab)^{-1} = b^{-1}a^{-1}$

If $(ab)^{-1} = b^{-1}a^{-1}$ then $ab = ba$

i) Let $ab = ba$

$$(ab)^{-1} = ((ab)^{-1})^{-1} = (b^{-1}a^{-1})^{-1}$$

$$(b^{-1}a^{-1})^{-1} = b^{-1}a^{-1}b^{-1}a^{-1}$$

Since a & b commute, we have:

$$b^{-1}a^{-1}b^{-1}a^{-1} = b^{-2}a^{-2}$$

ii) Let $(ab)^{-1} = b^{-1}a^{-1}$

$$\text{We know } (ab)^{-1} = ((ab)^{-1})^{-1} = b^{-1}a^{-1}b^{-1}a^{-1}$$

$$b^{-2}a^{-2} = (b^{-1})^{-1}(a^{-1})^{-1} = b^{-1}b^{-1}a^{-1}a^{-1}$$

For the two expressions to be equal a^{-1} & b^{-1} must commute

$$a^{-1}b^{-1} = b^{-1}a^{-1}$$

$$(a^{-1}b^{-1})^{-1} = (b^{-1}a^{-1})^{-1}$$

$$ba = ab$$

$$\therefore ab = ba$$